

DEREK WANG

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Education

High School

Expected Graduation: June 2025

IB Diploma — Ontario Secondary School Diploma

Bayview Secondary School

- IBDP Student — **IB Predicted: 40/45, Top 6 GPA: 95.8/100**
- Courses: HL English Lit, HL Economics, HL Chemistry, SL French B, SL Mathematics AA, SL Physics

Work Experience

City of Richmond Hill

March 2024 – Current

Youth Badminton Instructor

Rouge Woods Community Center, Richvale Community Center

- Coach a class of **14 beginner to intermediate students** (ages **9–16**) in **8-week terms**, focusing on footwork, technique, and strategy
- Designed a **new badminton curriculum** centered around **real game situations**, which was adopted by **all** badminton classes at Rouge Woods and Richvale, enhancing student engagement
- Provide personal **feedback and mentorship** while delivering lessons to improve **individual performance**, fostering an inclusive space

Extra Curricular Experiences

Checkmate Robotics Club | C++ | *Source Code*

September 2024 – Current

- Software team member of Team **16868C Rushdown** for the **VEX Robotics Competition**, invited to the **2025 VRC World Championships**
- Utilized **Okapi PROS Lib** to program robot functionalities, **Odometry** to track robot positioning, **PID Controllers** for precise motion control and error handling
- Wrote and tuned 15-second autonomous routes for matches, and 60-second programming skills, ranking **5th out of 80 teams** at **Provincial Championships**

Dr. X Robotics Academy | Java

July 2019 – May 2022

- Hardware team member of Team **16417 Dr. X Academy** for the **FIRST Tech Challenge**, invited to the **2022 FTC World Championships**
- Created **8 iterations** of the team's robot, including versions of the **drivetrain, arm, claw, turret** and a **backup robot**
- Write and tuned 30-second autonomous period for matches, using **java**, implementing **Dead Reckoning** for robot localization, and **PID Controllers** for motion control

Bayview Robotics Club | *Instagram* | *Github*

September 2021 – Current

- Led robotics team as **Vice President and main coder** in various robotics competitions such as **FIRST Tech Challenge** and **York Region Skills Challenge**
- Mentor **15 general members** and organize weekly STEM workshops, teaching them robotics basics in hardware and software.

Bayview Computer Club | *Instagram*

September 2024 – Current

- Plan and Deliver bi-weekly lessons on competitive programming **data structures and algorithms** to **20 members**, preparing them for the Canadian Computer Competition and future coding interviews

Projects

WebTyper | JavaScript | HTML | *Source Code*

- Created educational coding content for **75,000+** developers, with **4,200,000+** impressions per month
- Produced coding tutorials on advanced topics such as **data structures, algorithms, and software design principles**

Technical Skills

Languages: Java, C++, Python, HTML5, CSS, JavaScript

Developer Tools: Github

Libraries/Frameworks: ReactJS, OpenCV